

NEURAL BELLMAN OPERATORS

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We study the computation of economic policies when preferences, financial choices, and aggregate resource constraints enter the same dynamic problem. Neural Bellman Operators separate the approximation of continuation values from feasible policy improvement. We establish a finite-horizon value-loss bound that distinguishes evaluation error, improvement error, boundary error, and approximation of the transition operator. The bound supports two implementable safeguards: comparison with a feasible candidate set and convex improvement certified by a linear minimization gap. A stopped portfolio economy makes preference adjustment operational without introducing artificial wealth transfers. Its value is decreasing and convex in the adjustment-cost coefficient, and its optimal discounted adjustment expenditure is decreasing in that coefficient, even though individual policy coordinates need not be monotone. We implement neural policies and critics, evaluate their induced policies independently, and solve a coupled stochastic resource-allocation problem with binding shared constraints. A convex scenario-tree calculation bounds the latter policies' cost losses at held-out initial states. Recursive utility, sophisticated temporal selves, and dynamic capacity competition supply additional analytical and computational tests. The results identify which approximation errors matter economically and which conclusions require stronger information than a small training residual.

KEYWORDS: Dynamic programming, Neural policy iteration, Recursive utility, Endogenous preferences, Stochastic control, Computational economics.

1. INTRODUCTION

A computational policy is useful only to the extent that its approximation errors can be related to the decisions and welfare objects of the economic model. This requirement becomes demanding when controls affect both the distribution of future resources and the agent's preferences. An error in the continuation value can change consumption, portfolio risk, and the incentive to adjust preferences. A numerically feasible policy can also solve the wrong economy if the treatment of a state-space boundary implicitly adds resources. For recursive preferences and strategic interaction, even the object called a continuation value must be specified before an algorithm can be evaluated.

This paper develops Neural Bellman Operators (NBO) as a separated approximation of economic control operators. A critic represents a policy's continuation value. An actor proposes a feasible control using that continuation value. A distinct improvement calculation checks the proposal, and an independent evaluation measures the value of the resulting policy. We use both differential Hamiltonian operators and finite-step transition operators. The latter provide an executable route from a specified stochastic economy to neural training without requiring the critic's second derivatives. Their approximation of a continuous-time model is assessed separately from optimization of the finite-step problem.

The distinction between proposal and improvement is important. A stationary point of a neural actor's parameter objective need not maximize the Hamiltonian. An exact

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pointwise policy-improvement theorem therefore does not establish convergence of an unconstrained neural optimizer. We address this difference constructively. A candidate safeguard includes the actor’s proposal in a feasible comparison set. For convex control problems, a second safeguard uses a computable linear minimization gap to bound the remaining improvement error. Neither safeguard assumes that a small parameter gradient implies an economically optimal action.

Our first result gives an error account for finite-horizon policy value. It applies to monotone Bellman operators and permits a different approximation error at each date. The bound distinguishes the critic’s evaluation defect, the actor’s improvement gap, terminal or boundary data, and the discrepancy between the implemented and target transition operators. The theorem concerns functions and operators, not a particular optimizer. It becomes a numerical certificate only when its error bounds have been established on the required domain. This distinction allows ordinary held-out diagnostics and rigorous finite-problem certificates to appear in the same computational framework without being confused.

Our second result concerns endogenous preferences. In the portfolio economy, risk aversion is a state variable that can be adjusted at a quadratic cost. We specify a liquidation rule at the first exit from the operating domain. The model retains stochastic preference shocks, financial risk, their covariance, and all three controls. It does not replenish wealth when a simulated path crosses a lower boundary. For this economy, the value is the upper envelope of affine functions of the adjustment-cost coefficient. Consequently, the value is decreasing and convex in that coefficient, and optimal discounted cumulative adjustment effort is decreasing. The conclusion incorporates changes in portfolios, consumption, and the stopping time. It does not require, and does not imply, pointwise monotonicity of the preference-adjustment policy.

We implement the separated operator in two economically different environments. The portfolio exercise uses ordinary multilayer neural networks and a finite-candidate improvement safeguard. Its independent evaluation exposes substantial differences between initial-state performance and uniform accuracy near the liquidation boundary. A second application allocates investment across interacting stocks subject to a shared resource budget. It has correlated shocks, a nonseparable state cost, and binding constraints. A convex neural critic supports globally checked one-period improvement, while an independent nonanticipative scenario-tree optimization bounds policy cost losses at the specified initial states. This experiment supplies a dynamic, constrained comparison; it is not a static quadratic exercise relabeled as a scalability test.

The paper also retains recursive utility, temporal selves, dynamic games, nonsmooth solutions, and stochastic trace calculations. These extensions play different roles. Homothetic Merton and Epstein–Zin problems test actual evaluation and actor updates against independently derived policies. A sophisticated consumption problem tests the distinction between the current self’s criterion and ordinary continuation evaluation. A stochastic capacity game tests unilateral dynamic deviations, not just joint-profit maximization. Finally, counterexamples and trace calculations identify invalid shortcuts in neural approximation. These applications are not pooled into a single numerical success rate.

1.1. *Related Literature and the Contribution*

Policy improvement precedes neural approximation. [Jacka and Mijatović \(2015\)](#) give continuous-time policy-improvement results under explicit stochastic-control hypotheses. Our exact-operator proposition is a conditional statement within that tradition; its

purpose is to make the limiting argument and boundary requirements explicit. The new computational analysis concerns the errors that remain when evaluation and improvement are approximate.

The closest neural comparisons also combine value approximation and control improvement. [Kim, Kim, Kim, and Cho \(2025\)](#) use neural fixed-policy PDE evaluation within policy iteration. Their subsequent revision, [Kim, Kim, Kim, and Cho \(2026\)](#), develops interior error bounds and analyzes unresolved boundary information. [Lee and Kim \(2024\)](#) combine policy iteration with deep operator learning, including transfer across terminal functions. [Cai, Fang, and Zhou \(2025\)](#) train value and control representations using martingale conditions without requiring an explicit Hamiltonian optimizer. Direct approximation of stochastic controls by neural networks is already present in [Han and E \(2016\)](#). Thus neural evaluation, policy factorization, mesh-free training, and avoidance of a closed-form argmax are not claims of priority in this paper.

Our contribution is the combination of an explicit economic boundary contract, a value-loss account for the implemented operator, and improvement safeguards whose error has a direct interpretation. The finite-horizon bound is deliberately architecture-independent. The constrained stochastic experiment instantiates its actor component with a convex certificate, and the endogenous-preference result supplies an economic comparative static that does not rely on an unverified policy surface. These are specific claims rather than an assertion that a general combination of neural networks and dynamic programming is new.

The broader numerical literature includes projection and recursive methods ([Judd, 1998](#), [Stokey, Lucas, and Prescott, 1989](#)), sparse-grid approaches ([Brumm and Scheidegger, 2017](#)), and neural differential-equation solvers ([Raissi, Perdikaris, and Karniadakis, 2019](#), [Han, Jentzen, and E, 2018](#)). The relevant comparison depends on structure. A problem with an exact Riccati representation should be compared with that representation. A constrained convex dynamic program should be compared with a solver that exploits convexity. A general nonlinear problem requires its own reference or verification argument. We do not infer a universal dimension threshold from the cost of differentiating a network.

Convex neural parameterizations and optimization over their inputs have an established role in control; see [Amos, Xu, and Kolter \(2017\)](#). Our resource-allocation implementation uses a transparent convex feature class rather than proposing a new input-convex architecture. Its role here is to make the feasible improvement gap computable and to connect it to an independently bounded policy cost.

The following sections define the economic operator, establish approximation results, develop the preference economy, and report the computations. The final substantive section treats recursive utility, temporal selves, and dynamic games. A standalone supplement provides proofs, implementation details, additional diagnostics, and the conditions needed to interpret extensions beyond the executed experiments.

2. ECONOMIC CONTROL AND SEPARATED BELLMAN OPERATORS

2.1. *The Controlled Problem*

Let $S_t = (u_t, X_t)$ be a Markov state and let $a_t = (\theta_t, b_t)$ collect internal and external controls. On a specified domain D , the controlled state satisfies

$$dS_t = \mu(t, S_t, a_t) dt + B(t, S_t, a_t) dW_t, \quad a_t \in A(t, S_t), \quad (1)$$

where W is a vector of independent Brownian motions and $A(t, s)$ is the feasible correspondence. Correlation is represented in B . More generally, a continuous martingale with covariance density $Q(t, s, a)$ gives covariance $\sigma Q \sigma^\top$. A predictable covariance process not determined by the declared state does not, merely by notation, define the same Markov problem; the state must be augmented or the formulation changed.

For additive preferences and stopping time $\tau \leq T$, policy evaluation is

$$J^\pi(t, s) = \mathbb{E}_{t,s}^\pi \left[\int_t^\tau e^{-\rho(v-t)} r(v, S_v, \pi_v) dv + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right]. \quad (2)$$

The stopping rule and payoff are economic primitives. In an exit problem, τ is the first exit or terminal date. Reflection instead requires a regulator and a boundary operator involving its direction; state constraints require viability. Neither interpretation is supplied by an unreported numerical projection.

For recursive preferences, evaluation is first defined for each admissible policy by

$$-dY_t^\pi = f(t, S_t, \pi_t, Y_t^\pi) dt - Z_t^\pi dW_t, \quad Y_\tau^\pi = g(\tau, S_\tau). \quad (3)$$

The driver may incorporate a preference-adjustment cost. Existence, integrability, comparison, and any domain restriction on its value argument are imposed before taking a supremum over policies. Lipschitz drivers on a finite horizon provide a standard admissible class; the Epstein–Zin specification below uses its own positive-domain and transversality conditions. Calling Y an argument of utility does not itself establish well-posedness.

When a classical Markov representation is available, write

$$\mathcal{H}^a[v] = f(t, s, a, v) + \mu(t, s, a)^\top \nabla v + \frac{1}{2} \text{tr}\{B(t, s, a)B(t, s, a)^\top \nabla^2 v\}. \quad (4)$$

For additive utility, $f = r - \rho v$. The evaluation equation is $v_t + \mathcal{H}^\pi[v] = 0$ with the specified boundary operator, while the control equation replaces $\mathcal{H}^\pi$ by $\sup_{a \in A} \mathcal{H}^a$. Boundary data, admissibility, and comparison are part of these statements, not consequences of a neural parameterization.

2.2. Evaluation, Proposal, and Improvement

Let v_ξ be a critic and a_ω a feasible actor. A differential evaluation step freezes the policy and minimizes

$$L_E(\xi; \pi) = \mathbb{E}_v \left[(\partial_t v_\xi + \mathcal{H}^\pi[v_\xi])^2 \right] + \lambda_B \mathbb{E}_{v_B} \left[|\mathcal{B}v_\xi - g_B|^2 \right]. \quad (5)$$

Here $\mathcal{B}$ is the actual boundary operator. A Dirichlet discrepancy is appropriate for the stopped problem used below, not for an unspecified reflected diffusion. Hard architectures can impose the boundary exactly, although they do not eliminate interior approximation error.

The actor’s Hamiltonian proposal solves, approximately,

$$\max_{\omega} \mathbb{E}_{v_I} \left[\mathcal{H}^{a_\omega} \{ \text{sg}(v_\xi), \text{sg}(\nabla v_\xi), \text{sg}(\nabla^2 v_\xi) \} \right]. \quad (6)$$

The notation sg freezes the critic’s parameter block. Coefficients depending on the action, including diffusion loadings, retain their action derivatives. Evaluation does not

update actor parameters, and actor improvement does not update critic parameters. For a game, rival controls are also frozen in a player’s improvement step. These restrictions concern the computational graph, not just the labels attached to two loss functions.

A bounded activation can enforce a box constraint. Shared resource constraints require a different map. For example, $a_i = B_0 \exp(z_i) / (\sum_{j=1}^d \exp(z_j) + \exp(z_0))$ gives $a_i \geq 0$ and $\sum_i a_i \leq B_0$, with z_0 representing slack. General convex sets can use projection; a penalty alone is an approximation whose violations must be measured. An arbitrary policy sampling measure does not turn (6) into the gradient of lifetime welfare. Such an interpretation requires occupancy or adjoint weighting and the relevant interchange conditions.

For an implemented transition rule, it is often preferable to use a finite-step operator. At dates $n = 0, \dots, N - 1$, define

$$(\mathcal{T}_n^a v)(s) = R_n(s, a) + \int v(s') K_n(ds' | s, a), \quad \mathcal{T}_n v = \sup_{a \in A_n(s)} \mathcal{T}_n^a v. \quad (7)$$

The nonnegative kernel includes discount and survival. A stopped payoff is included in R_n ; it is not continued at a clipped interior state. The experiments with nonlinear critics proceed backward: improve the actor against the frozen next-date critic, then fit the current critic to the evaluated target of the selected action. This is a finite-horizon separated Bellman update, not a simulation of a joint scalar loss.

In this semigroup implementation, freezing the continuation network’s parameters must not remove its derivative with respect to the next state. The actor influences s' through the transition, so the composition $v_{n+1}(F(s, a, \varepsilon))$ must retain that input derivative. Our tests compare this actor gradient with finite differences and separately check the differential Hamiltonian graph. The distinction prevents a stop-gradient convention appropriate for a Hamiltonian jet from being incorrectly imposed on a finite-step transition.

2.3. The Exact Policy-Improvement Benchmark

The following proposition states the exact object to which neural approximation is compared. Let $\mathcal{V}$ be a classical value space whose norm controls $v, v_t, \nabla v, \nabla^2 v$, and the boundary traces required by $\mathcal{B}$. Including the time derivative is essential to pass the evaluation equation to a limit.

ASSUMPTION 1—Exact evaluation and improvement: *The policy class Π is compact in a stated uniform topology. Evaluation $E : \Pi \rightarrow \mathcal{V}$ is uniquely defined, continuous, and has precompact image. A selector I attains the pointwise Hamiltonian supremum, maps Π into itself, and is continuous at accumulation points of the policy sequence. Comparison and verification hold for the specified generator, recursive driver, boundary operator, and value class. Evaluated values are bounded above by the optimal value.*

PROPOSITION 1—Exact separated policy improvement: *Under Assumption 1, let $\pi_{j+1} = I(\pi_j)$ and $v_j = E(\pi_j)$. The evaluated values increase to a common limit in $\mathcal{V}$. Every accumulation policy p satisfies $E(p) = E(I(p)) = \bar{v}$, and $\bar{v}$ solves the maximized boundary-value problem. If comparison identifies the value, p is optimal.*

The important limit is equality of values, not equality of successive policies. Along $\pi_{j_k} \rightarrow p$, continuity gives $E(p) = \bar{v}$ and $E(I(p)) = \bar{v}$ from the shifted sequence. The

evaluation equation for $I(p)$ at that same value is the maximized equation. The supplement proves this step in detail. Compactness and closure under an exact selector are substantial hypotheses; they are not automatically properties of a finite neural class. Proposition 1 is not an optimizer convergence theorem.

3. APPROXIMATION ERROR AND FEASIBLE IMPROVEMENT

3.1. An Error Account for Policy Value

Let $\mathcal{T}_n^a$ be the target evaluation operator and $\tilde{\mathcal{T}}_n^a$ the implemented operator. For additive preferences, these have the form (7). For recursive preferences, the following result also permits nonlinear evaluation maps, provided they are order preserving and Lipschitz in the continuation value. We impose those properties on the relevant value domain rather than presume them for every recursive aggregator.

ASSUMPTION 2—Finite-horizon operators: *For every feasible a , the target operator $\mathcal{T}_n^a$ is order preserving and satisfies $\|\mathcal{T}_n^a v - \mathcal{T}_n^a w\|_\infty \leq \gamma_n \|v - w\|_\infty$. The same bound holds for $\tilde{\mathcal{T}}_n^a = \sup_a \mathcal{T}_n^a$; the supremum is well defined. Terminal values and the selected policy values exist and are bounded. The numerical errors defined below are finite. The constants γ_n may exceed one for a finite-horizon recursive problem.*

For learned values $\hat{v}_n$ and a feasible policy π_n , define

$$\epsilon_n = \left\| \hat{v}_n - \tilde{\mathcal{T}}_n^{\pi_n} \hat{v}_{n+1} \right\|_\infty, \quad (8)$$

$$\delta_n = \left\| \tilde{\mathcal{T}}_n \hat{v}_{n+1} - \tilde{\mathcal{T}}_n^{\pi_n} \hat{v}_{n+1} \right\|_\infty, \quad (9)$$

$$\kappa_n = \sup_{s,a} \left| \tilde{\mathcal{T}}_n^a \hat{v}_{n+1}(s) - \mathcal{T}_n^a \hat{v}_{n+1}(s) \right|, \quad \eta = \|\hat{v}_N - g_N\|_\infty. \quad (10)$$

The first quantity is evaluation error; the second is a payoff improvement gap, not an action distance. The third compares operators on the actual learned continuation. It includes transition, quadrature, interpolation, and stopped-payoff errors when those are present. Let $\Gamma_{n,j} = \prod_{i=n}^{j-1} \gamma_i$, with an empty product equal to one.

THEOREM 2—Policy-value error account: *Under Assumption 2, the target policy value J_n^π satisfies*

$$\|\hat{v}_n - J_n^\pi\|_\infty \leq \sum_{j=n}^{N-1} \Gamma_{n,j} (\epsilon_j + \kappa_j) + \Gamma_{n,N} \eta. \quad (11)$$

For the target optimal value V_n^* ,

$$0 \leq V_n^*(s) - J_n^\pi(s) \leq \sum_{j=n}^{N-1} \Gamma_{n,j} (2\epsilon_j + \delta_j + 2\kappa_j) + 2\Gamma_{n,N} \eta. \quad (12)$$

The proof is backward error propagation, but the placement of the errors matters. The policy-evaluation recursion contributes $\epsilon_n + \kappa_n$. The optimized recursion contributes $\epsilon_n + \delta_n + \kappa_n$. Adding the resulting bounds yields (12); separately inflating an actor

gap by operator error would produce an unnecessarily loose coefficient. For additive stopped control with exact discount, $\gamma_n \leq e^{-\rho \Delta_n}$. A boundary-payoff error at an intermediate exit belongs in the relevant one-step operator error, not only in the terminal error at N .

There is an analogous differential statement. If a classical candidate v has boundary discrepancy at most η , fixed-policy residual at most $\epsilon(t)$, and Hamiltonian gap at most $\delta(t)$, Itô's formula and verification give

$$V^*(t, s) - J^\pi(t, s) \leq 2\eta + \int_t^T e^{-\rho(q-t)} \{2\epsilon(q) + \delta(q)\} dq. \quad (13)$$

The stopped martingale and comparison conditions are stated in Supplement S.1. This is a quantitative link to a differential critic, not a claim that sampled residuals provide the required supremum bounds.

A finite test set by itself provides a lower observation of a supremum, not an upper certificate. With a known Lipschitz constant L and an h -cover, a residual's uniform magnitude is bounded by its largest tested magnitude plus Lh . Without such regularity and coverage information, we report held-out errors as diagnostics. The same distinction applies to critic derivatives, boundary layers, and off-policy states. Network approximation of a smooth function does not convert a sample average into a uniform error bound.

3.2. Two Improvement Safeguards

Let $Q_n(s, a) = \tilde{T}_n^a \hat{v}_{n+1}(s)$. For a finite feasible set $C_n(s)$ and neural proposal $a_\omega(s)$, the candidate safeguard selects

$$\pi_n(s) \in \arg \max_{a \in C_n(s) \cup \{a_\omega(s)\}} Q_n(s, a). \quad (14)$$

It weakly improves the proposal's evaluated target at every queried state. If $C_n(s)$ is an h_a -cover of the feasible set and Q_n has a verified action-Lipschitz constant L_a , its remaining improvement gap is at most $L_a h_a$. Without that information, a finite comparison is not a continuum optimality certificate. In the portfolio computation we use the safeguard and report the remaining independently evaluated feasible deviations.

For convex cost minimization, a stronger check is available. Let $F_s(a)$ be differentiable and convex on a compact convex feasible set $A(s)$. Define

$$g_s(a) = \nabla F_s(a)^\top a - \min_{b \in A(s)} \nabla F_s(a)^\top b. \quad (15)$$

Convexity implies $0 \leq F_s(a) - \min_b F_s(b) \leq g_s(a)$. For $A = \{a \geq 0 : \sum_i a_i \leq B_0\}$, the linear minimum is $B_0 \min\{0, \min_i \partial_i F_s(a)\}$. The computation therefore obtains an improvement certificate without a global search over neural parameters. The actor is a proposal; projected improvement is stopped according to (15).

In the resource application, the continuation critic is convex by construction, so the one-period criterion is strongly convex in the action. Its neural representation uses nonnegative coefficients on squared-ReLU features together with a convex quadratic term and a free affine term. This architecture justifies the convex check for the *learned* continuation criterion. It does not assume that its continuation value is exact. An

independent scenario-tree calculation separately assesses the resulting policy's lifetime cost.

PROPOSITION 3—Finite stochastic-program policy certificate: *Let $C(u)$ be the convex cost of a nonanticipative finite scenario-tree control problem on a compact convex feasible set $\mathcal{U}$. Let $\hat{u}$ be a feasible reference point, $G(\hat{u})$ its gap defined as in (15), and u^π the feasible tree induced by an evaluated feedback policy. Then*

$$C(u^\pi) - C(\hat{u}) \leq C(u^\pi) - C^* \leq C(u^\pi) - C(\hat{u}) + G(\hat{u}). \quad (16)$$

This certificate is independent of the training residual. It applies at each initial state whose complete scenario tree is evaluated. It is not a bound for an untested continuum of initial conditions, nor does it cover a different shock distribution. The supplement gives the projected-simplex implementation and the independent tree gradient.

3.3. Stationarity and Nonsmooth Solutions

A smooth Hamiltonian can have a strictly suboptimal attracting stationary point. On $[-2, 2]$, consider

$$H(a) = -(a^2 - 1)^2 + 0.2a. \quad (17)$$

Its negative local maximizer is approximately -0.973994 , while its global maximizer is approximately 1.024120 . The former is locally attracting under gradient ascent and has an improvement gap of approximately 0.399875 . Exact evaluation of $V^a(t) = (T - t)H(a)$ does not change that fact. Isolation of stationary points cannot turn an actor-ODE limit into an optimal policy. An optimality conclusion needs, for example, a justified global concavity or gradient-dominance condition together with the necessary representation and evaluation assumptions. Constant-step Adam and finite-step L-BFGS are not covered by such a conclusion merely because they update separate parameter blocks.

Nor does a small differential residual select a viscosity solution. For the exit equation $1 - |v'| = 0$ on $(-1, 1)$ with zero boundary values, the relevant value is $|x| - 1$. The functions

$$w_\varepsilon(x) = \sqrt{1 + \varepsilon^2} - \sqrt{x^2 + \varepsilon^2} \quad (18)$$

have the same boundary values and squared residual tending to zero, but converge to the wrong function $1 - |x|$. At zero, that function fails the viscosity subsolution inequality. Our regression suite retains this example and the biased composite-loss example in Supplement S.2. A convergent approximation of a nonsmooth control problem needs an appropriate mechanism, such as monotonicity, stability, consistency, boundary treatment, and comparison (Barles and Souganidis, 1991); it is not supplied by smooth activation functions alone.

3.4. The Squared-Trace Objective

For a differential critic, let $A = B^\top \nabla^2 v B$ and let z_k be independent standard Gaussian probes. Conditional on a fixed state, action, and parameter vector, put $q_k = z_k^\top A z_k$, $\bar{q} = K^{-1} \sum_k q_k$, and $r_k = a - q_k/2$. Then

$$\mathbb{E}(a - \bar{q}/2)^2 = (a - \text{tr}(A)/2)^2 + \frac{\|A\|_F^2}{2K}. \quad (19)$$

Averaging r_k^2 instead has an excess expectation independent of K . These are different objectives, even though each individual trace estimate is unbiased.

PROPOSITION 4—An unbiased squared-residual estimand: For $K \geq 2$, define

$$U_K = \frac{(\sum_{k=1}^K r_k)^2 - \sum_{k=1}^K r_k^2}{K(K-1)}. \quad (20)$$

Then $\mathbb{E}U_K = (a - \text{tr}(A)/2)^2$. If differentiation and expectation can be interchanged, its parameter gradient is unbiased for the gradient of that squared residual.

The estimator can be negative in a batch. Unbiasedness does not establish low variance, stability of its minimization, or an accurate policy. Optimizing over actions using the same noisy probes also introduces a selection issue beyond the fixed-action identity. We therefore retain exact derivatives for the differential benchmark checks and assess the probe estimands with repeated independent batches. The executed high-dimensional resource solver uses a finite-step operator and a convex critic; its results are not attributed to Hutchinson acceleration.

4. ENDOGENOUS PREFERENCES AND PORTFOLIO CHOICE

4.1. A Stopped Operating Economy

We now specify an economy in which the preference-adjustment margin is active in the transition law. Wealth is X , risk aversion is u , and the controls are consumption c , preference adjustment θ , and risky-asset share π . Inside $D = (1.2, 2.8) \times (0.5, 2.0)$, the state evolves as

$$du_t = \theta_t dt + \sigma_u dZ_t^u, \quad (21)$$

$$dX_t = \{(r + \pi_t(\mu - r))X_t - c_t\} dt + \pi_t X_t \sigma_X dZ_t^X, \quad d\langle Z^u, Z^X \rangle_t = \varrho dt. \quad (22)$$

The operating contract terminates at $\tau = T \wedge \inf\{t : S_t \notin D\}$. At that time the agent receives the cardinal payoff

$$G(u, X) = -0.02(u - 2)^2 + 0.1 \log X. \quad (23)$$

The payoff summarizes liquidation or the outside option after leaving the operating regime; it is not a transfer restoring the state to the interior.

The feasible box is $c \in [0.05, 0.8]$, $\theta \in [-0.2, 0.2]$, and $\pi \in [-0.5, 0.8]$. We set

$$T = 1, \quad \rho = 0.04, \quad r = 0.02, \quad \mu = 0.08, \quad \sigma_X = 0.2, \quad \sigma_u = 0.05, \quad \varrho = -0.25. \quad (24)$$

The baseline adjustment-cost coefficient is $k = 2$. For a feasible policy,

$$J_k^\pi(t, u, X) = \mathbb{E}_{t, u, X}^\pi \left[\int_t^\tau e^{-\rho(s-t)} \left\{ \frac{c_s^{1-u_s}}{1-u_s} - \frac{k}{2} \theta_s^2 \right\} ds + e^{-\rho(\tau-t)} G(u_\tau, X_\tau) \right]. \quad (25)$$

Consumption is measured in the fixed units appearing in (25). An additive or multiplicative normalization depending on u changes incentives once u is controlled. We therefore do not treat such transformations as innocuous changes of utility representation.

This boundary contract is a substantive specification. Constant preference volatility cannot remain in a closed interval without an exit or regulator mechanism. Moreover, under the displayed consumption bounds, the drift at $X = 0.5$ is negative even at the largest feasible portfolio share: its maximum is -0.016 . An unregulated state-constrained interpretation is consequently unavailable. Stopping repairs that problem without modifying the interior diffusion, imposing an unreported reflection, or injecting wealth. It is not equivalent to the earlier clipped transition.

The stopped problem has bounded payoffs and Lipschitz coefficients before exit. A useful check is that G is a strict supersolution for every feasible control. The bounds in (24) imply

$$\frac{c^{1-u}}{1-u} - \frac{k\theta^2}{2} + \mathcal{L}^{c,\theta,\pi}G - \rho G \leq -0.8162 \quad (k \geq 0). \quad (26)$$

The supplement derives the bound without a grid maximization. Together with a fixed nondegenerate admissible diffusion, it supplies a boundary-continuity argument for the stopped value. The comparison also has economic content: continued operation has a cost relative to the liquidation payoff. Exit incentives are therefore part of this calibration and must be reported, rather than interpreted as a generic wealth or risk-aversion effect.

4.2. The Operator and Control Margins

Writing subscripts for derivatives, the HJB equation in the interior is

$$\begin{aligned} -V_t = \sup_{c,\theta,\pi} \{ & \frac{c^{1-u}}{1-u} - \frac{1}{2}k\theta^2 - \rho V + \theta V_u + [(r + \pi(\mu - r))X - c]V_X \\ & + \frac{1}{2}\sigma_u^2 V_{uu} + \frac{1}{2}\pi^2 X^2 \sigma_X^2 V_{XX} + \varrho \sigma_u \sigma_X \pi X V_{uX} \}, \end{aligned} \quad (27)$$

with $V = G$ on the spatial boundary and at T . Both diagonal diffusion terms and the mixed term belong to the operator. In particular, the generator of u^2 contains σ_u^2 even if the preference drift is zero.

For an interior optimum, the first-order conditions are

$$c^{-u} = V_X, \quad k\theta = V_u, \quad 0 = (\mu - r)XV_X + \pi X^2 \sigma_X^2 V_{XX} + \varrho \sigma_u \sigma_X X V_{uX}. \quad (28)$$

When $V_X > 0$, $V_{XX} < 0$, and all relevant controls are interior, these give

$$c = V_X^{-1/u}, \quad \theta = V_u/k, \quad \pi = -\frac{(\mu - r)V_X}{X\sigma_X^2 V_{XX}} - \frac{\varrho \sigma_u V_{uX}}{X\sigma_X V_{XX}}. \quad (29)$$

The final expression separates a mean-variance term from a preference-hedging term. Its sign is not determined by $\varrho < 0$ alone: the cross derivative and curvature also matter. Nor need the stopped economy satisfy $V_X > 0$ throughout the domain. The constrained implementation compares feasible controls rather than mechanically applying these formulas where their premises fail. At a binding box constraint, the appropriate one-sided derivative or complementary-slackness condition replaces an interior equality.

The cardinal utility derivative is

$$\partial_u \left(\frac{c^{1-u}}{1-u} \right) = \frac{c^{1-u} \{1 - (1-u) \log c\}}{(1-u)^2}. \quad (30)$$

It need not have the sign suggested by a verbal interpretation of increased risk aversion. The full shadow value V_u incorporates this derivative, future opportunities, adjustment costs, covariance, and liquidation. The numerical policy is informative only relative to these declared primitives.

4.3. A Comparative Static That Survives Endogenous Policies

Fix an initial state and let

$$C^\pi = \mathbb{E}^\pi \int_0^\tau e^{-\rho t} \frac{\theta_t^2}{2} dt, \quad B^\pi = J_k^\pi + kC^\pi. \quad (31)$$

For a fixed policy, B^π and C^π are independent of k : changing the coefficient does not change the state law unless the policy is changed. In particular, the stopping time is allowed to be endogenous to the policy.

THEOREM 5—Adjustment costs and cumulative effort: *Suppose the feasible policy class and all primitives other than k are fixed, and optimizers exist at the coefficients being compared. Then $V(k) = \sup_\pi \{B^\pi - kC^\pi\}$ is decreasing and convex. If $k_2 > k_1$ and π_i is optimal at k_i , then*

$$C^{\pi_2} \leq C^{\pi_1}. \quad (32)$$

Each $-C^{\pi_i}$ is a subgradient of V at k_i . At differentiability points, all optimal effort levels equal $-V'(k)$. For policies with objective losses at most e_1, e_2 at the respective coefficients,

$$C^{\pi_2} - C^{\pi_1} \leq \frac{e_1 + e_2}{k_2 - k_1}. \quad (33)$$

The proof uses the two optimality inequalities and is given in Supplement S.3. The approximate version is useful when numerical policies are used in an economic comparison: the effort ordering can be assessed against a welfare-error budget. The result does not hold pointwise for $\theta(t, u, X)$ merely because (29) contains $1/k$. The shadow value, portfolio choice, consumption, and exit distribution all change when the problem is re-solved.

The theorem is an economic comparative static, not an identification result. Inferring k from data additionally requires an observation model for latent preferences, an identified moment map, and control of numerical error in that map. The solver supplies policy and value objects for such a model; a graph of the policy alone cannot establish identification. This distinction leaves the structural-estimation motivation intact while identifying the additional objects it would require.

5. COMPUTATION AND INDEPENDENT POLICY EVALUATION

5.1. The Stochastic Transition and Its Boundary

For the portfolio economy we use four equiprobable weak-Euler branches. Independent signs $z_1, z_2 \in \{-1, 1\}$ generate

$$\Delta u = \theta h + \sigma_u \sqrt{h} z_1, \quad \Delta X = [(r + \pi(\mu - r))X - c]h + \pi X \sigma_X \sqrt{h} \{ \varrho z_1 + \sqrt{1 - \varrho^2} z_2 \}. \quad (34)$$

TABLE I
JOINT STATE-TIME REFINEMENT OF THE STOPPED PREFERENCE MODEL

Grid	Dates	V_0	C_0	Seconds
17×25	4	-1.140548	0.009344	0.31
33×49	8	-1.136916	0.010335	2.07
65×97	16	-1.135019	0.010902	15.45

Initial state $(2, 1.25)$; $k = 2$. All rows optimize $5 \times 5 \times 7$ consumption, adjustment, and portfolio actions. C_0 is discounted integrated $\theta^2/2$. These are finite-approximation values, not certified continuous-time errors.

Thus both variances and the cross covariance are present. On each branch we follow the line segment from s to $s + \Delta s$ until its first boundary intersection, at fraction $\alpha \in [0, 1]$ of the step. The branch contribution is

$$r_k(s, a) \frac{1 - e^{-\rho \alpha h}}{\rho} + e^{-\rho \alpha h} \{1_{\{\alpha=1\}} v_{n+1}(s + \Delta s) + 1_{\{\alpha < 1\}} G(s + \alpha \Delta s)\}. \quad (35)$$

An initial boundary state stops immediately and receives G . An interior step that hits the boundary exactly may equivalently use its boundary datum. No branch is returned to an interior state after an exit.

This polygonal stopped random walk is an approximation to Brownian stopping, not an exact Brownian first-passage simulation. The reference operator uses positive bilinear interpolation for live continuations. The neural training operator evaluates its frozen continuation network at those same physical successor locations. Policy evaluation subsequently uses a separate NumPy implementation with positive interpolation. These two approximations are compared rather than assumed to be identical.

Generator tests use $1, u, X, u^2, X^2, uX$. At an interior test point, their discrepancies have the expected drift-squared Euler remainder and recover the complete covariance matrix. A separate test verifies that opposite preference controls produce different conditional means even when the displacement is smaller than a grid cell. Interpolation preserves that difference; nearest-node rounding can erase it. The differential actor graph, finite-step actor graph, and immediate stopping boundary are also tested independently.

Table I reports a refinement sequence with $\max(\Delta u, \Delta X)^2/h$ decreasing from 0.04 to 0.01. This matters because interpolation error divided by the time step must be controlled in a generator approximation. The action mesh is held fixed in this sequence, and action refinement is assessed separately. These calculations diagnose discretization sensitivity; three meshes do not themselves constitute an error certificate for the continuous-time value.

5.2. Preference-Cost Comparisons

The cost panel solves the entire finite problem again at $k = 0.5, 2, 8$, using a 25×37 state grid, eight dates, and $9 \times 9 \times 13$ controls. Consumption, preference adjustment, and portfolio share are all optimized. Discounted effort is evaluated under the resulting policy, including its effect on the exit time. Table II reports the initial state $(u, X) = (2, 1.25)$.

Discounted cumulative effort falls as k increases, consistently with Theorem 5. The portfolio and consumption coordinates at this initial state are at their bounds for all three displayed coefficients. This is not evidence for the interior formulas in (29).

TABLE II
RE-SOLVED PREFERENCE-ADJUSTMENT COST PANEL

k	V_0	C_0	c_0	θ_0	π_0
0.5	-1.117524	0.014614	0.800	0.200	-0.500
2.0	-1.135835	0.009888	0.800	0.200	-0.500
8.0	-1.160291	0.001432	0.800	0.100	-0.500

Each row re-solves all dates and all three control coordinates on the same 25×37 grid, eight dates, and $9 \times 9 \times 13$ action mesh. The initial state is $(2, 1.25)$. The comparison concerns cumulative effort, not a universal pointwise policy ordering.

In particular, a short portfolio position in this calibration should not be described as an unconditional preference-hedging effect. The liquidation contract, the level of the running utility, and the portfolio's effect on exit jointly determine the optimum. Occupation-weighted exit probabilities are computed by forward propagation of the interpolation-defined Markov chain; an unweighted fraction of exiting grid branches would measure a different object.

5.3. Neural Policies in the Preference Economy

The ordinary neural proposal has two tanh hidden layers of width 32 and three bounded outputs. The safeguarded critic has two tanh hidden layers of width 64. Its output is $G(s) + (1 - t)m(s)N_{\xi}(s)$, where m vanishes on every face of the rectangle. The implementation uses separate L-BFGS fits, 2,048 uniformly sampled training states per seed, eight dates, and a feasible $5 \times 5 \times 7$ comparison set containing the actor's proposal. No grid reference values or optimal reference policies enter these training targets.

After training, the entire feedback sequence is frozen and evaluated on a 33×49 grid. The comparison reference uses the same number of dates and a finer $9 \times 9 \times 13$ action mesh. A separate $7 \times 7 \times 11$ deviation search uses the *evaluated policy continuation*, not the learned critic, and includes the incumbent action. Consequently, a reported gain is a feasible payoff improvement. It is not a norm of the difference between two actions and is not an upper bound over the continuum of actions.

Table III reports all three safeguarded seeds and retains a lower-budget, unsafeguarded pilot. The pilot differs in width and optimization budget, so its comparison is a development diagnostic, not an isolated causal ablation. Table IV reports errors for the full control vector. A small error at the initial state does not imply a small maximum error near the boundary, and a policy coordinate can change substantially near a switch. The critic, policy value, and feasible-deviation columns therefore remain distinct.

The hard boundary architecture attains the displayed boundary condition to numerical precision. It does not certify the interior value or the Brownian stopping approximation. The sizable worst-case diagnostics identify where additional computation is needed for a uniform guarantee; they are not suppressed by averaging across states or seeds. Conversely, the initial-state comparison and the re-solved cost panel provide economic information that is not captured by the largest training residual alone.

TABLE III

NEURAL POLICIES AND INDEPENDENT EVALUATION IN THE STOPPED ECONOMY

Run	$V_R - J_0$	Critic RMSE	Critic max.	Gain max.	Seconds
Pilot 101	0.026720	0.066694	0.676974	1.340226	13.5
Safe 101	0.001018	0.047925	0.298592	0.228318	88.3
Safe 202	0.001366	0.044197	0.346920	0.210235	83.0
Safe 303	0.001177	0.052661	0.934824	0.228568	81.5

V_R is the separately solved 33×49 , eight-date, $9 \times 9 \times 13$ reference at $(2, 1.25)$. Critic errors compare the initial critic with its independently evaluated policy over the whole grid. Gain is the largest feasible one-step improvement over all interior states and dates against the evaluated continuation. The pilot has a smaller critic and training budget; it is not a matched ablation. Hard boundary discrepancies are evaluated separately in the raw record.

TABLE IV

FULL-VECTOR POLICY DISCREPANCIES AGAINST THE MATCHED REFERENCE

Run	MAE c	MAE θ	MAE π	Max. c	Max. θ	Max. π
Pilot 101	0.0136	0.0678	0.3212	0.562	0.400	1.300
Safe 101	0.0047	0.0165	0.1065	0.562	0.400	1.300
Safe 202	0.0047	0.0157	0.0954	0.562	0.400	1.300
Safe 303	0.0048	0.0158	0.1085	0.562	0.400	1.300

Mean absolute and maximum discrepancies include every interior common node and all eight dates. Coordinates have different economic units and are not added together or called a Hamiltonian gain. Both policies use their stated finite approximations; these are not errors against a continuous-time oracle.

5.4. Coupled Stochastic Resource Allocation

To examine a genuinely dynamic shared constraint, let $x_n \in \mathbb{R}^d$ denote productive stocks and $a_n \in \mathbb{R}_+^d$ investment. The three-period model is

$$x_{n+1} = A_d x_n + a_n + e_{n+1}, \quad A_d = 0.8I + 0.04(I_{+1} + I_{-1}), \quad \sum_i a_{ni} \leq 0.25d. \quad (36)$$

Here I_{+1}, I_{-1} denote the adjacent off-diagonal matrices. The shock has four equally likely realizations, $e = z_1 \ell_1 + z_2 \ell_2$, where $\ell_{1i} = 0.06$ and ℓ_{2i} is evenly spaced from -0.04 to 0.04 . Its covariance is rank two and generally has nonzero cross-stock entries. Heterogeneous initial stocks, neighbor spillovers, a common budget, and the aggregate cost couple the state coordinates.

The objective is expected total cost,

$$C^\pi(x) = \mathbb{E}_x^\pi \left[\sum_{n=0}^2 \left\{ q(x_n) + \frac{0.1}{d} \|a_n\|^2 \right\} + q(x_3) \right], \quad (37)$$

$$q(x) = \frac{\|x - \mathbf{1}\|^2}{d} + 0.2 \left(\frac{\mathbf{1}^\top (x - \mathbf{1})}{d} \right)^2.$$

This is a finite stochastic economy, not a discretization claim about a different diffusion. All 64 terminal shock paths are used when a policy is evaluated. The independent reference optimizes a nonanticipative scenario tree with 21 decision nodes per initial

TABLE V
COUPLED STOCHASTIC RESOURCE ALLOCATION AT A COMMON ACCURACY TARGET

d	NBO upper loss	Proposal only	NBO sec.	Reference sec.	Passed
4	0.000645	0.01190	1.737	0.172	3/3
8	0.000461	0.09541	2.241	0.190	3/3
16	0.000567	0.06807	3.180	0.226	3/3

Loss columns are worst held-out cost-loss upper bounds across three seeds and 32 initial states per seed. Proposal only removes deployment improvement from the same fitted actor weights. Timing is the median across seeds at a 10^{-3} cost-loss target; NBO includes fitting and full evaluation, and the reference solves the same initial-state batch to its 10^{-3} gap target. A separate 10^{-7} reference gap is used for auditing, not for the timing comparison. The reference is a convex nonanticipative scenario tree, not a sparse grid.

state. Its objective is convex quadratic and its nodewise resource constraints are genuine restrictions on investment.

The cost critic is

$$\hat{C}_n(x) = q(x) + b_n + \ell_n^\top(x - \mathbf{1}) + \sum_{j=1}^J w_{nj} [h_j^\top(x - \mathbf{1}) + b_j]_+^2, \quad w_{nj} \geq 0. \quad (38)$$

The fixed hidden features include coordinate, aggregate, and random directions; $J = 6d + 102$. Output coefficients are fitted by constrained least squares to the separated Bellman targets. A width-32 tanh proposal network uses a slack-softmax feasibility map and is trained against the improved controls. At deployment, projected convex improvement uses the analytic critic gradient, checked against automatic differentiation, and the gap (15). Squared-ReLU features are suitable here because the finite-step operator needs a value and an action gradient, not a classical second-derivative residual.

For each of $d = 4, 8, 16$ and seeds 101, 202, and 303, training uses 1,024 draws from $[-0.25, 1.65]^d$. Evaluation uses 32 fresh initial states from $[0, 1.2]^d$ and their complete shock trees. The reference is solved both to the common cost tolerance 10^{-3} for timing and to a gap below 10^{-7} for independent validation. Formula (16) gives a numerical upper bound on each tested policy’s cost loss. These are double-precision convexity certificates; recorded feasibility and gradient discrepancies accompany them, rather than an assertion of interval-arithmetic proof.

Every displayed run meets the common cost-error target at its tested initial states. The shared constraint binds frequently, including at most or all initial states depending on dimension and seed. Thus the experiment tests a dynamically consequential budget rather than an inactive constraint. The same trained proposal networks are also evaluated without the deployment safeguard. This matched ablation, reported in the supplement, isolates the effect of the improvement calculation without changing network weights or training data.

At this query batch and tolerance, the specialized convex reference is faster end to end than training and evaluating the neural representation. The latter delivers a reusable feedback approximation, but a crossover in amortized cost requires a separate measurement. The experiment establishes accurate constrained neural control for this model and a transparent cost-accuracy comparison; it does not establish superiority over a solver that exploits the model’s convex structure.

5.5. Smooth and High-Dimensional Controls

An additional stochastic LQR experiment uses correlated Gaussian shocks, neighbor coupling in the transition, and a rank-one cross-state cost at dimensions 4, 8, and 16. Its critic has quadratic features, its actor is linear, and its independent reference is Riccati recursion. This structured control checks backward evaluation, covariance contributions, actor coefficients, and full-policy values. It is not an experiment with a generic multilayer critic, nor a substitute for the constrained resource application.

Computational cost depends on the operator and representation. For a differential critic and rank- r loading B , the exact trace can be evaluated using r directional Hessian contractions. A K -probe trace instead uses K such contractions after forming the probe directions. Their total costs include network differentiation, loading multiplication, action optimization, sampling, and the iterations needed to achieve an error target. Per-iteration differentiation does not establish a polynomial solution-complexity theorem, and no threshold such as dimension 50 follows from that calculation. The semigroup experiments report their own costs without attributing them to unexecuted stochastic-Hessian acceleration.

6. RECURSIVE UTILITY, TEMPORAL SELVES, AND DYNAMIC GAMES

6.1. Merton and Epstein–Zin Policy Evaluation

The smooth portfolio benchmark uses stationary infinite-horizon CRRA utility with $\rho = 0.04$, $\gamma = 2$, $\mu = 0.08$, $r = 0.02$, and $\sigma = 0.2$. With $c = mX$ and no terminal bequest, the optimal ratios are

$$\pi^* = \frac{\mu - r}{\gamma\sigma^2} = 0.75, \quad m^* = \frac{\rho + (\gamma - 1)\{r + (\mu - r)^2 / (2\gamma\sigma^2)\}}{\gamma} = 0.04125. \quad (39)$$

These are stationary quantities, not finite-horizon policies with an arbitrary terminal condition. Under the displayed optimum, the discounted negative wealth moment in the verification argument decays at rate $m^* > 0$.

For recursive utility, put $a = 1 - 1/\psi$ and $b = 1 - \gamma$. The difference-form Epstein–Zin driver on $c > 0$, $bV > 0$ is

$$f_{EZ}(c, V) = \frac{\rho}{a} \{c^a (bV)^{1-a/b} - bV\}. \quad (40)$$

The logarithmic case requires a limit rather than substitution at $a = 0$. In the numerical regime $\gamma = 5$, $\psi = 1.5$, with the same asset and discount parameters, homothetic optimization gives

$$\pi^* = 0.3, \quad m^* = \psi\rho + (1 - \psi)\{r + (\mu - r)^2 / (2\gamma\sigma^2)\} = 0.0455. \quad (41)$$

For $V = AX^b/b$, the coefficient is $A = (\rho(m^*)^{a-1})^{b/a}$. Consumption utility is increasing on the declared domain. A negative value when $b < 0$ is required by that domain and is not a sign error.

We execute fixed-policy evaluation for the unknown coefficient A and separate automatic-differentiation updates for bounded consumption and portfolio parameters. The coefficient is represented by $A = \exp(q)$; it and the controls are learned, not set to

TABLE VI
EXECUTED HOMOTHETIC POLICY EVALUATION AND IMPROVEMENT

Model	Learned m	Learned π	Policy max. error	Relative A error
Merton	0.04124996	0.74999998	3.94e-08	9.13e-13
Epstein–Zin	0.04550010	0.30000000	1.03e-07	6.89e-12

The coefficient $A = \exp(q)$ and bounded actor parameters are learned by separated updates. Analytical values are used only to compute the displayed errors. The sign condition and normalized HJB are tested separately. These runs use homothetic features rather than an unrestricted neural value class.

their targets. The analytical ratios are used afterward for validation. Table VI distinguishes these homothetic-feature solves from the ordinary neural networks used for endogenous preferences.

Supplement S.4 derives the policy equations, the positive branch of the evaluation equation, and a verification class with the required transversality. For the displayed recursive optimum, the linearized discount rate is 0.1115, the relevant wealth moment grows at rate 0.066, and their difference is $0.0455 > 0$. A separate finite-horizon architecture test imposes $bV > 0$ and an exact terminal sign by a transformed output. It is a domain test, not an additional solution of a finite-horizon Epstein–Zin problem.

The role of these calculations is to test recursive value dependence and the evaluation graph. Neither a low residual nor the positivity transform is by itself a proof of recursive-utility existence for every policy. Equally, the fact that an argmax has no analytic formula does not define a fully nonlinear PDE. Semilinearity concerns dependence on the highest-order derivatives. Control of diffusion can make the HJB fully nonlinear, whereas a difficult drift-control maximization with a fixed diffusion need not do so. The FBSDE literature also contains value representations rather than exclusively Pontryagin adjoints (Han, Jentzen, and E, 2018); these methods should not be excluded by that distinction.

6.2. Sophisticated Temporal Selves

Let the state obey $s_{n+1} = F(s_n, c_n, \varepsilon_{n+1})$ and let $\delta = e^{-\rho\Delta}$. A sophisticated current self evaluates a one-period deviation using $\beta\delta$ on ordinary continuation utility. The equilibrium system is

$$c_n^*(s) \in \arg \max_{c \in \mathcal{C}(s)} \{u(c)\Delta + \beta\delta \mathbb{E}[V_{n+1}(F(s, c, \varepsilon))]\}, \quad (42)$$

$$V_n(s) = u(c_n^*(s))\Delta + \delta \mathbb{E}[V_{n+1}(F(s, c_n^*(s), \varepsilon))], \quad V_N = G. \quad (43)$$

The factor β belongs in the current self's improvement criterion but not in each step of the ordinary continuation evaluation. Repeating it in (43) instead defines a different discount recursion. We use the same equations in the main text, supplement, and implementation.

The executed example has four consumption dates, terminal log wealth, no interest, $w_{n+1} = w_n - c_n$, $0 < c_n < w_n$, and $\Delta = 1$. With $V_n(w) = A_n \log w + B_n$ and $A_N = 1$, backward induction gives

$$z_n \equiv \frac{c_n}{w_n} = \frac{1}{1 + \beta\delta A_{n+1}}, \quad A_n = 1 + \delta A_{n+1}, \quad B_n = \log z_n + \delta \{A_{n+1} \log(1 - z_n) + B_{n+1}\}. \quad (44)$$

TABLE VII

SOPHISTICATED CONSUMPTION RATIOS AND INDEPENDENT DEVIATIONS

β	c_0/w	c_1/w	c_2/w	c_3/w	Eval. residual
0.7	0.2827991	0.3401848	0.4312698	0.5978885	1.78e-15
1.0	0.2163112	0.2651939	0.3467520	0.5099987	1.78e-15

Four consumption dates and terminal log wealth, $\Delta = 1$, $\rho = 0.04$. Continuation is evaluated without multiplying by β again. Independent bounded one-shot optimizations against these frozen continuation values give no gain exceeding 10^{-9} on the tested wealth levels.

Strict concavity establishes the one-shot optimum. We compare it with independent bounded scalar optimization and evaluate the ordinary recursion at four wealth levels. All consumption columns are ratios, not levels. The no-present-bias case $\beta = 1$ recovers exponential discounting.

A two-consumption-date example with terminal log wealth and $\delta = 1$ makes the distinction transparent: the sophisticated first ratio is $1/(1 + 2\beta)$, whereas the erroneous geometrically repeated-beta recursion gives $1/(1 + \beta + \beta^2)$. They equal 0.416667 and 0.456621 at $\beta = 0.7$. The discrepancy is economic, not a change of notation.

6.3. Stochastic Capacity Competition

For player i , evaluation freezes the policy profile, and improvement maximizes only player i 's criterion with rivals' controls detached. A dynamic equilibrium must also survive unilateral changes in the player's entire continuation policy. We therefore distinguish a one-period Hamiltonian gap from dynamic exploitability,

$$\text{Exploit}_i(s) = \sup_{\tilde{\pi}_i} J_i(\tilde{\pi}_i, \pi_{-i}; s) - J_i(\pi_i, \pi_{-i}; s). \quad (45)$$

The executed game has two firms, binary capacities $K_i \in \{0, 1\}$, and demand $D \in \{1, 1.3\}$ with transition matrix $\begin{pmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{pmatrix}$. Output is $q_i = K_i/3$, and revenue is $q_i(D - q_i - q_j)$. Investment $a_i \in [0, 1]$ is the probability of capacity one next period, conditional on which the firms' capacity draws are independent. Its cost is $0.4a_i^2/2$. This includes the cost of maintaining capacity; it is not a free reset. The horizon is four dates, the discount factor is 0.95, and terminal value is $0.05K_i$.

For every date and all eight states, stage best responses are clipped affine functions of the rival's investment. Their composed contraction is checked, and iteration is repeated from three initial profiles. After the equilibrium profile is computed, a separate backward best-response program freezes the rival at every date and optimizes the deviating firm's continuation. The largest recorded dynamic gain is below the stated 10^{-10} tolerance. The saved policies and values cover the complete finite state space.

This is an independent dynamic-game reference, not a trained high-dimensional game network. A separate automatic-differentiation test verifies that the own-action gradient is present and the rival gradient is absent. The static Cournot comparison, with Nash quantities $1/3$ and joint-profit quantities $1/4$, remains a diagnostic for the distinction between equilibrium and collusion, not a substitute for the dynamic check. Supplement S.5 sets out the equilibrium operator, resource normalization for larger markets, and a conditional small-player approximation bound. Symmetry of initialization is not used as an equilibrium certificate.

The portfolio and recursive-utility benchmarks build on [Merton \(1971\)](#), [Epstein and Zin \(1989\)](#), and [Duffie and Epstein \(1992\)](#). The normalization and admissible comparison class used here are specified in Supplement S.4.

7. CONCLUSION

The economic content of a numerical policy depends on more than its training objective. Evaluation, feasible improvement, stochastic transition, boundary treatment, and equilibrium deviations are distinct objects. Neural Bellman Operators make their interfaces explicit and permit approximation errors to be related to policy value.

The finite-horizon error account shows how to combine critic, actor, boundary, and operator errors without identifying a small sampled residual with optimality. Feasible candidate comparison and convex improvement provide two ways to turn a neural proposal into a checked economic decision. The constrained stochastic resource experiment implements the latter and obtains independent numerical bounds on policy cost loss. The preference economy implements the former while retaining all internal and external controls, correlated shocks, and a complete liquidation contract. Its comparative-static theorem identifies a monotone aggregate effort object even when individual policy coordinates do not move monotonically.

Recursive utility, temporal selves, and dynamic games require their own continuation and admissibility conventions. Their analytical and computational tests support these extensions without conflating an algebraic target, a trained representation, and an equilibrium certificate. This combination of an economic model, an implemented operator, and an explicit error account is the basis on which additional dimensions, richer preferences, or structural estimation can be developed and evaluated.

7.0.0.1. Replication. The reviewable source is `ECTA_R4.tex`; the standalone supplement is `SUPP_R4.tex`. The replication directory contains the solvers, fixed configurations, raw results, trained weights, checkpoint replay checks, and deterministic table generator. A manifest links executed source hashes to a reachable source commit and separates execution status from numerical tolerances. Timing is retained as meta-data but excluded from the scientific-result digest. The archive preserves the previous manuscripts, embedded arrays, and advisory reports unchanged; historical illustrations do not enter the current tables. The accompanying response locates each substantive repair without making an assertion about an editorial decision.

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